PULMO AI OFFLINE PULMONOLOGY INTELLIGENCE – MK-XI

ADVANCED PLEURAL FLUID LAB

Query -> parse -> interpret -> graph -> PDF · all offline

Generated	10/15/2026, 6:49:17 PM
Engine	ASK parser + 30-disease Bayesian (Light 6e LRs) + 9-graph suite
Confidence	96%

QUERY ASKED

58M fever and pneumonia, purulent fluid, pH 7.08, glucose 25, LDH 1800, neutrophils 90%, gram positive

PLEURAL FLUID PANEL ENTERED

PARAMETER	VALUE
Appearance	Frank pus
Neutrophils (%)	90
PF glucose (mg/dL)	25
PF pH (blood-gas)	7.08
PF LDH (U/L)	1800
Cytology	Not sent
Gram / culture	Positive

INTERPRETATION FLAGS

Frank pus — empyema — drain, no further tests needed to decide

Neutrophil-predominant — acute process: parapneumonic, PE, pancreatitis, subphrenic abscess, early TB (< 2 wk symptoms)

Glucose < 60 mg/dL — narrow to: parapneumonic, malignant, TB, rheumatoid (+ paragonimiasis, hemothorax, Churg-Strauss, urinothorax, occasionally lupus)

Glucose < 40 mg/dL — drainage indicated if parapneumonic; RA often < 30, even < 5

pH < 7.20 — the 10 causes: complicated PPE, esophageal rupture (lowest, < 6.0), RA, TB, malignant, hemothorax, systemic acidosis (check blood pH), paragonimiasis, lupus, urinothorax

LDH > 1,000 U/L — intense inflammation: empyema, RA, esophageal rupture, malignancy; serial rising LDH = escalate workup

Gram stain positive — large bacterial burden — immediate drainage regardless of chemistry

Rheumatoid pleuritis: the most extreme chemistry: glucose < 30, pH < 7.2, LDH > 1,000 — confirm RF >= 1:320

Complicated parapneumonic: neutrophils + pH/glucose hit — classify (Light/ACCP) and drain

GUARDRAIL ALERTS

EMPYEMA — DRAIN NOW

Frank pus or positive Gram stain = immediate tube thoracostomy (small-bore 9–12 F acceptable, irrigate q6h). No further chemistry needed to decide.

Check blood pH

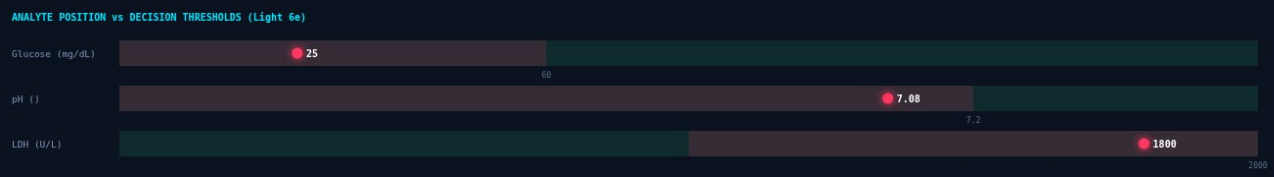
When PF pH < 7.20, a simultaneous arterial pH is required to exclude systemic acidosis as the cause.

BAYESIAN DIFFERENTIAL (LEADING 8)

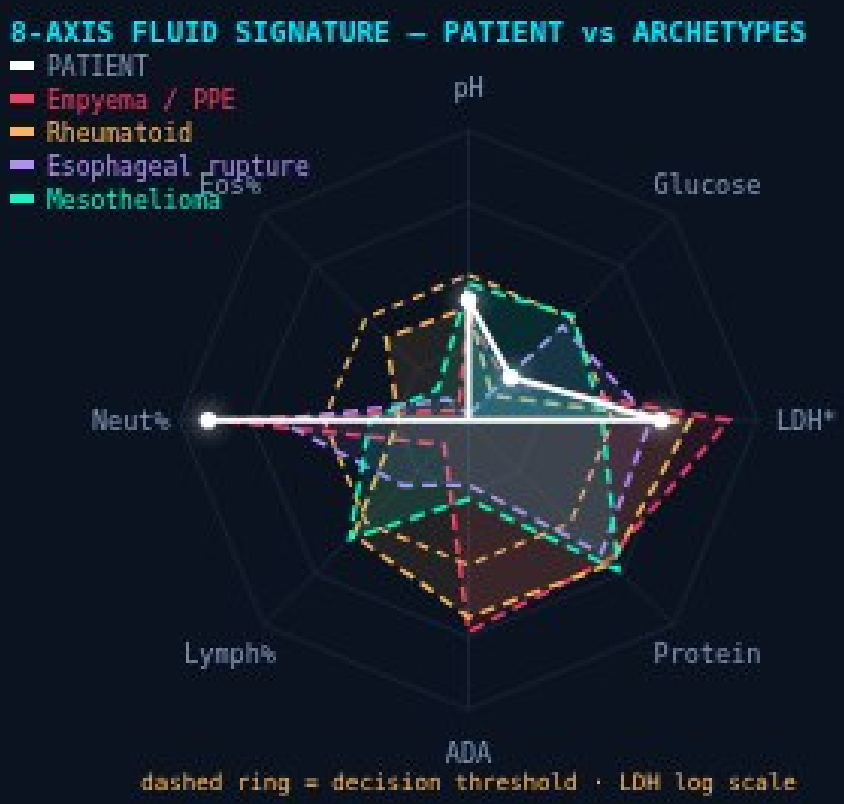
DIAGNOSIS	POSTERIOR
Parapneumonic effusion / empyema	99.9%
Rheumatoid pleuritis	0.0%
Esophageal rupture (Boerhaave)	0.0%
Peritoneal dialysis effusion	0.0%
Malignant mesothelioma	0.0%
Urinothorax	0.0%
Subphrenic / intra-abdominal abscess	0.0%
Drug-induced pleural effusion	0.0%

GRAPHS

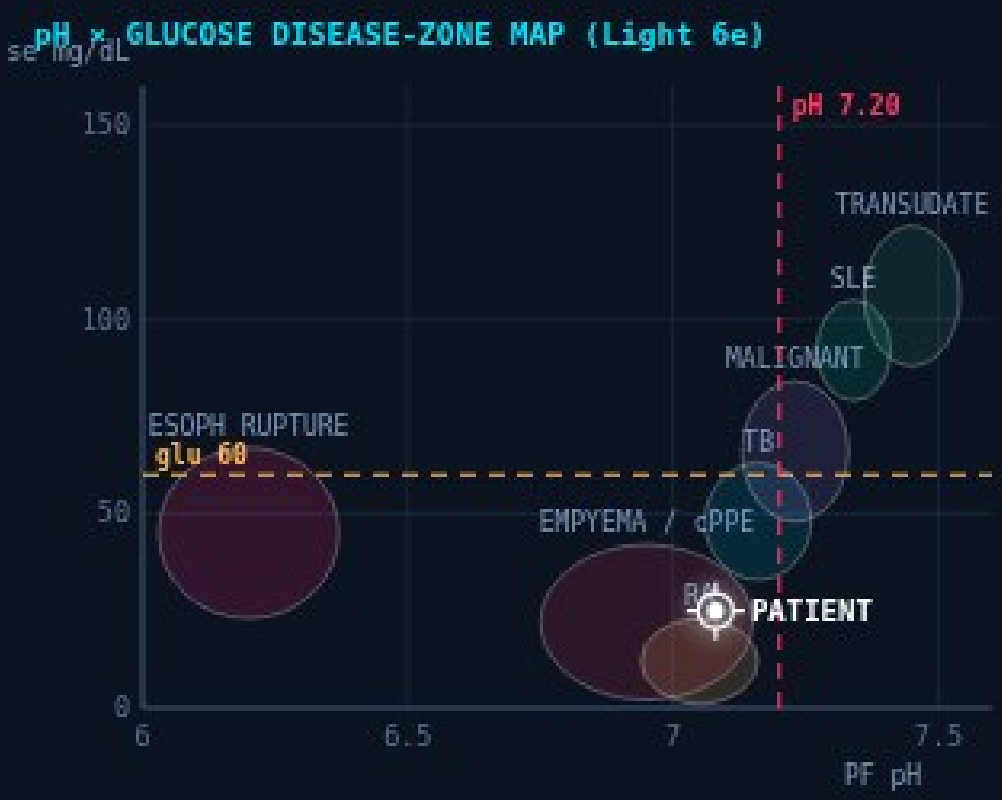
ANALYTE POSITION vs DECISION THRESHOLDS



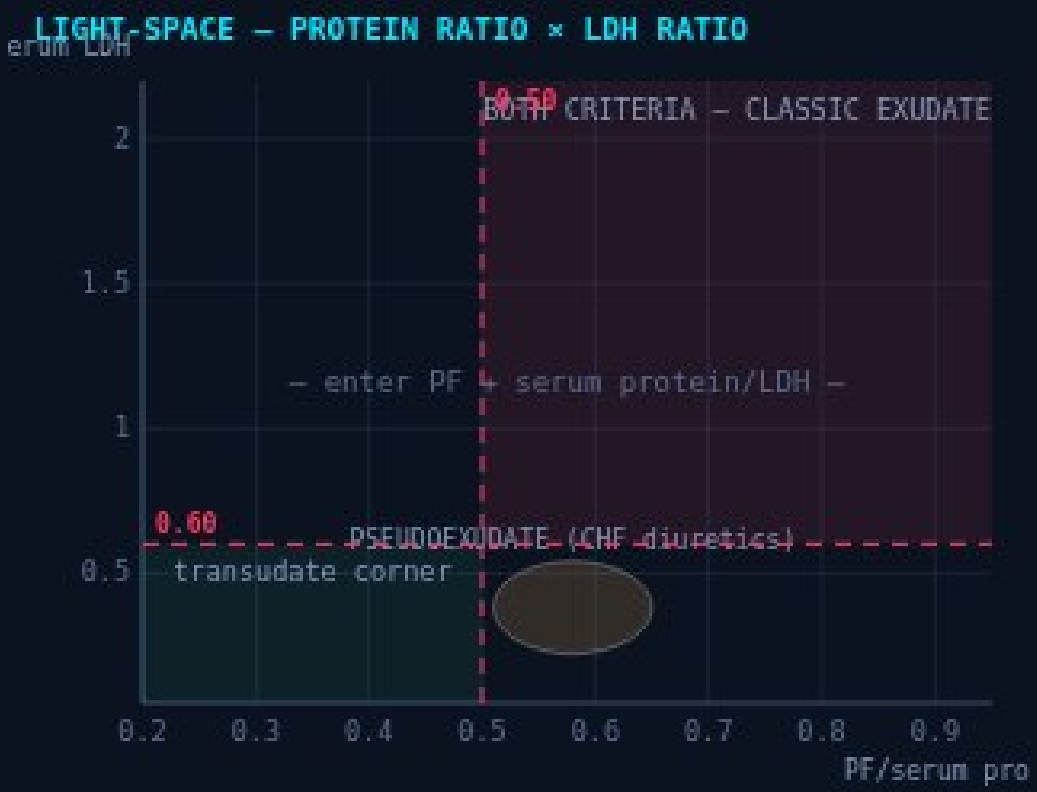
8-AXIS FLUID SIGNATURE — PATIENT vs DISEASE ARCHETYPES



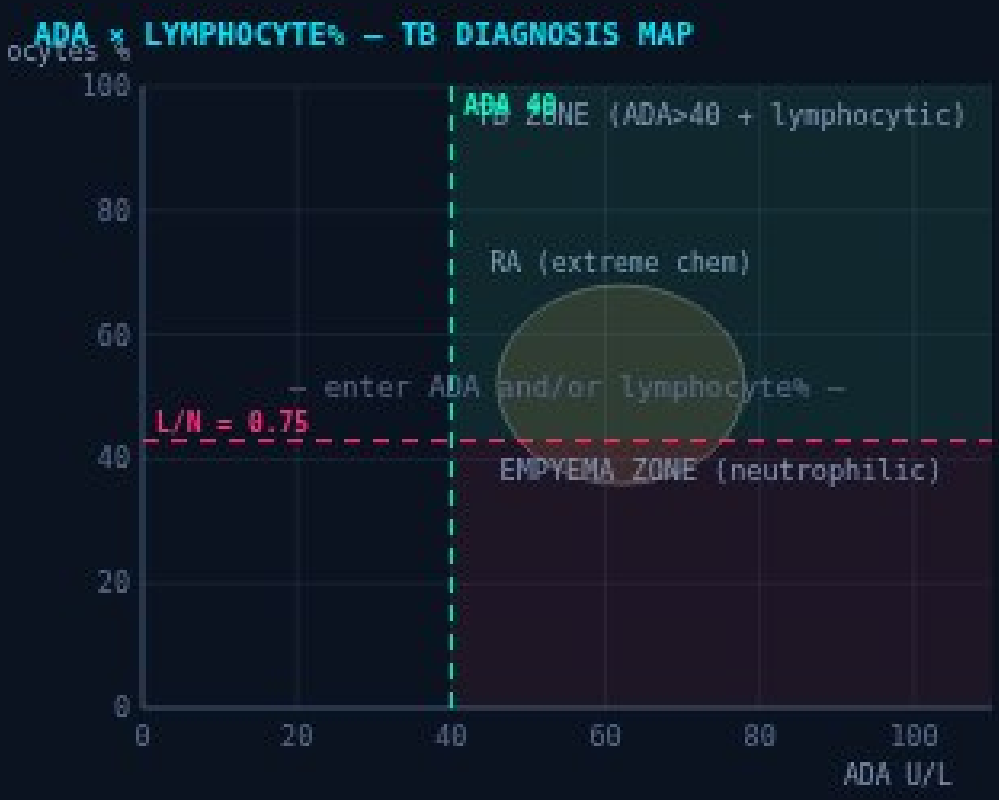
DISEASE-ZONE MAP — pH × GLUCOSE (Light 6e)



LIGHT-SPACE — PROTEIN RATIO × LDH RATIO



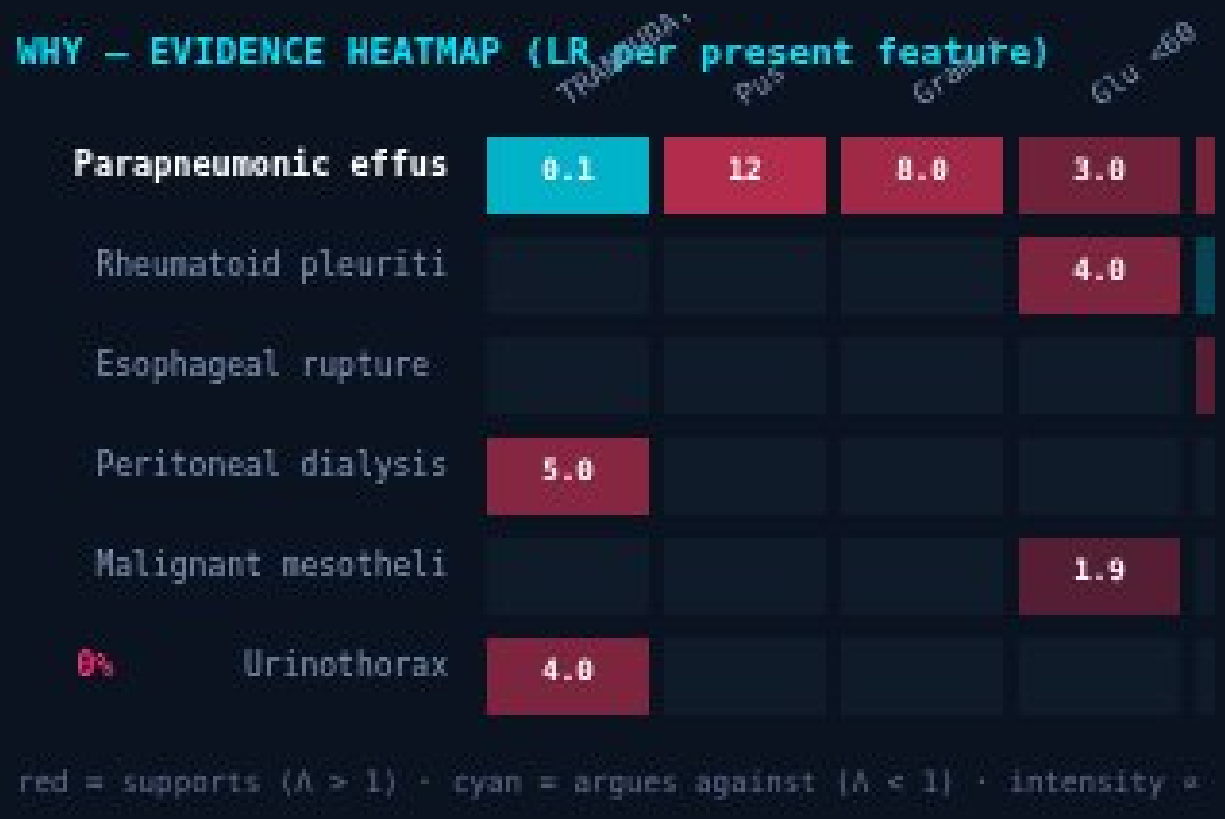
TB DIAGNOSIS MAP — ADA × LYMPHOCYTE %



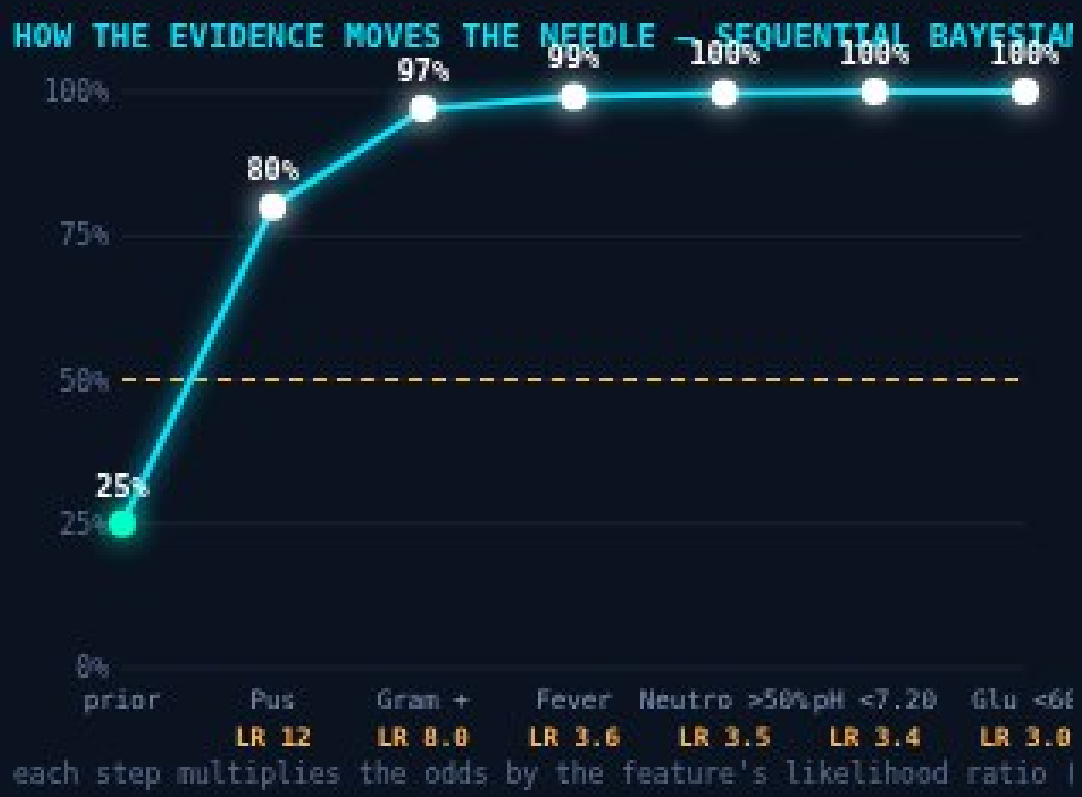
BAYESIAN DIFFERENTIAL — 30 DISEASE MODELS



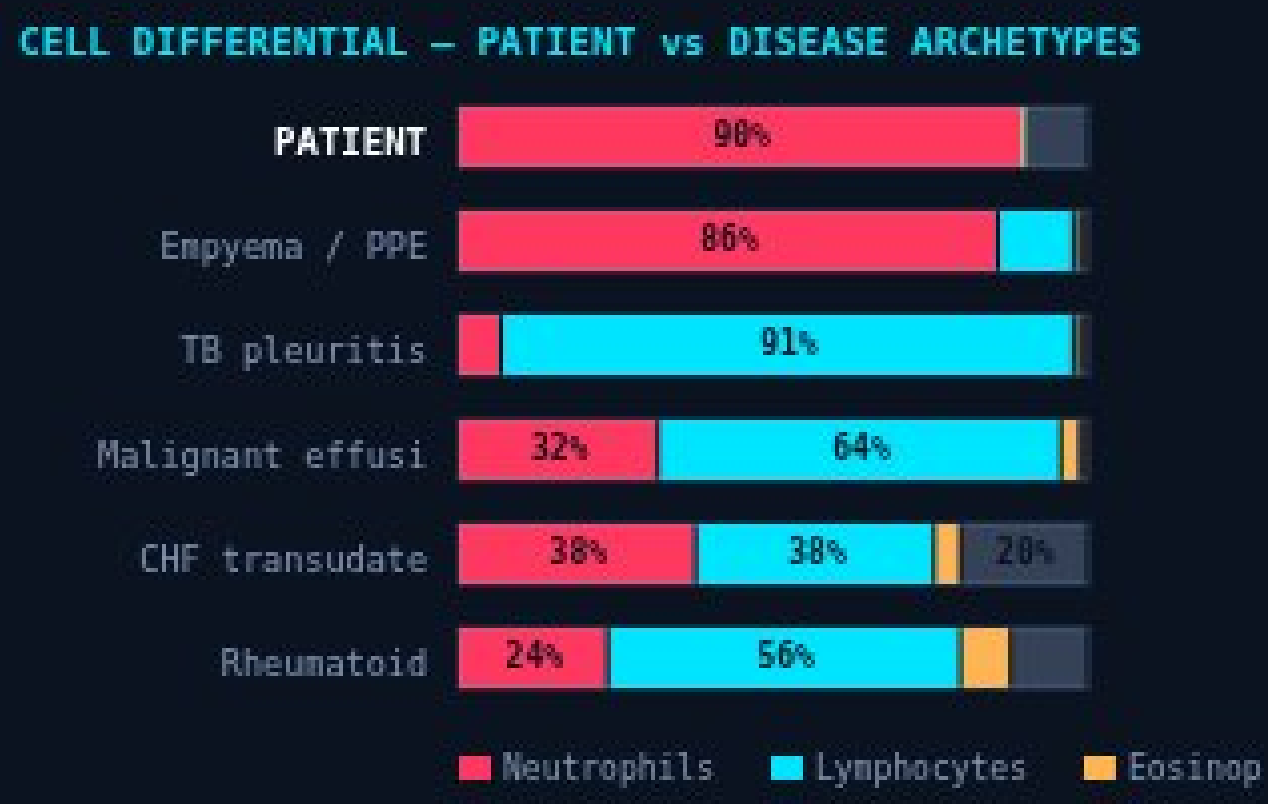
EVIDENCE HEATMAP — LIKELIHOOD RATIOS PER FEATURE



BAYESIAN LADDER — SEQUENTIAL PROBABILITY UPDATES



CELL DIFFERENTIAL STACKS — PATIENT vs ARCHETYPES



NEXT DIAGNOSTIC STEPS (DATA-DRIVEN LADDER)

- Diagnostic thoracentesis (> 10 mm on decubitus/US): send protein + LDH (paired serum!), cells, glucose, pH, cytology, TB marker — the single highest-yield test.

MANAGEMENT — PARAPNEUMONIC EFFUSION / EMPYEMA

- Antibiotics per pneumonia syndrome (add anaerobic cover if aspiration/feculent)
- Class 3 (pH 7.00–7.20): antibiotics + serial thoracentesis
- Class 4/5 (pH < 7.00, glucose < 40, Gram+, loculated): tube thoracostomy 9–12 F + q6h saline irrigation if small-bore
- Intrapleural tPA 10 mg + DNase 5 mg BID × 3 days (MIST2) if loculated/poor drainage
- VATS decortication if not improving after ~10 days or lung fails to expand
- Necrotizing/gas-forming organisms: early surgical consultation

Corpus: Light's Pleural Diseases 6th Edition (Richard W. Light) — full 522-page scan (native text + OCR). Thresholds, likelihood ratios and disease archetypes transcribed from chapters 7–8 and disease chapters.